

Module 3: CloudWatch Alarms

Assignment

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Problem Statement: You work for XYZ Corporation. To maintain the security of the AWS account and the resources you have been asked to implement a solution that can help easily recognize and monitor the different users. Also, you will be monitoring the machines created by these users for any errors or misconfigurations.

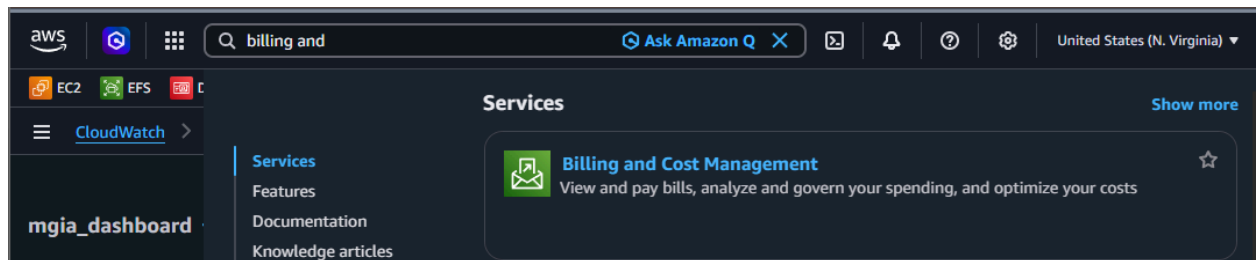
Tasks To Be Performed: The AWS account and its resources were monitored using Amazon CloudWatch. A billing alarm was configured to monitor estimated AWS charges, and an EC2 CPU utilization alarm was configured to monitor the performance of an EC2 instance and send notifications through Amazon SNS.

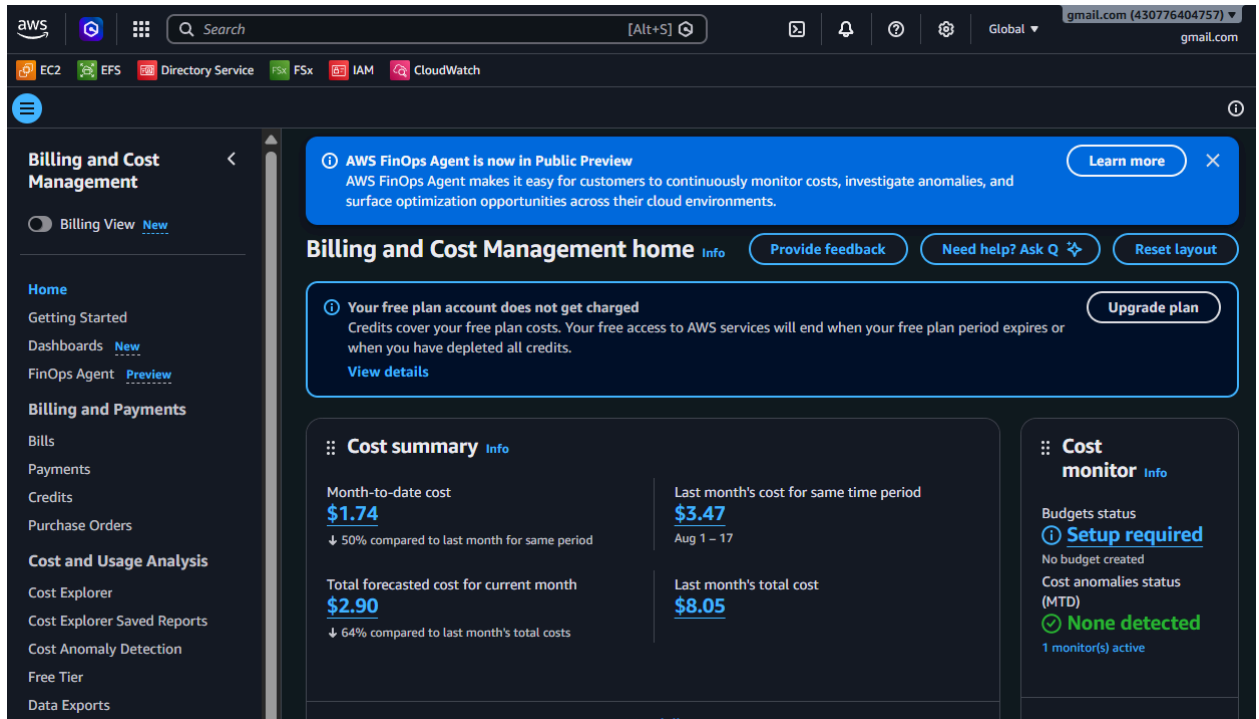
1. To create a CloudWatch billing alarm which goes off when the estimated charges go above \$500.
2. To create a CloudWatch alarm which goes off to an Alarm state when the CPU utilization of an EC2 instance goes above 65%. Also add an SNS topic so that it notifies the person when the threshold is crossed.

Enabling billing alerts.

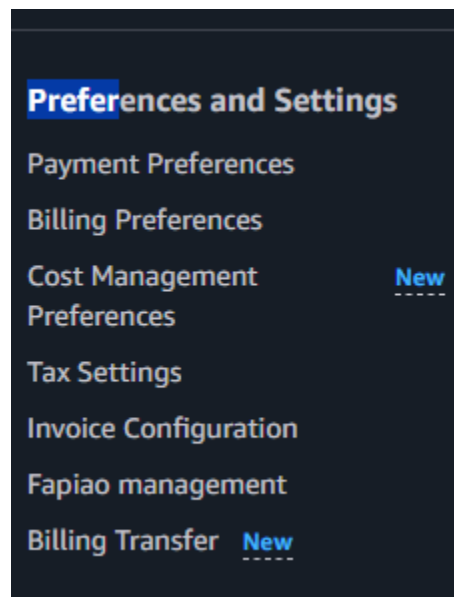
AWS requires billing alerts to be enabled before the billing metric becomes available.

1. Opened **Billing and Cost Management**.

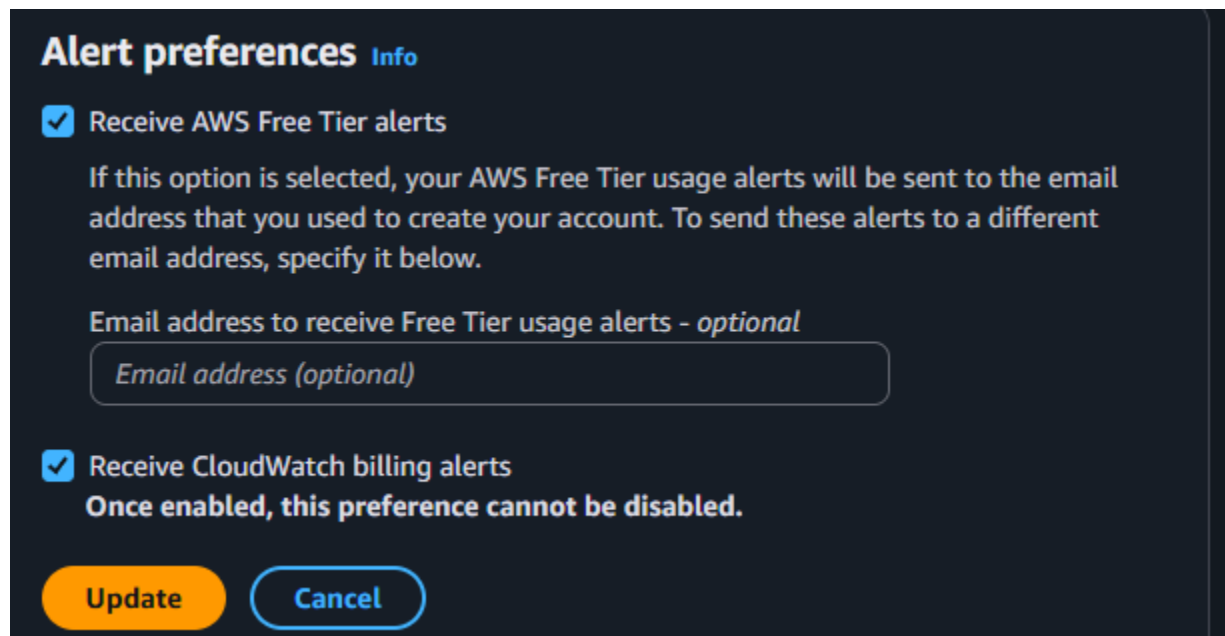
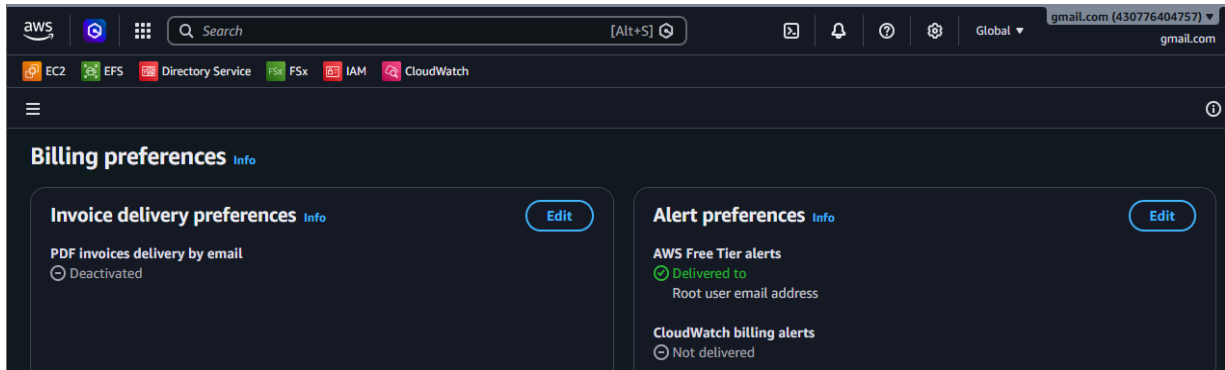




2. Navigated to **Billing Preferences**.



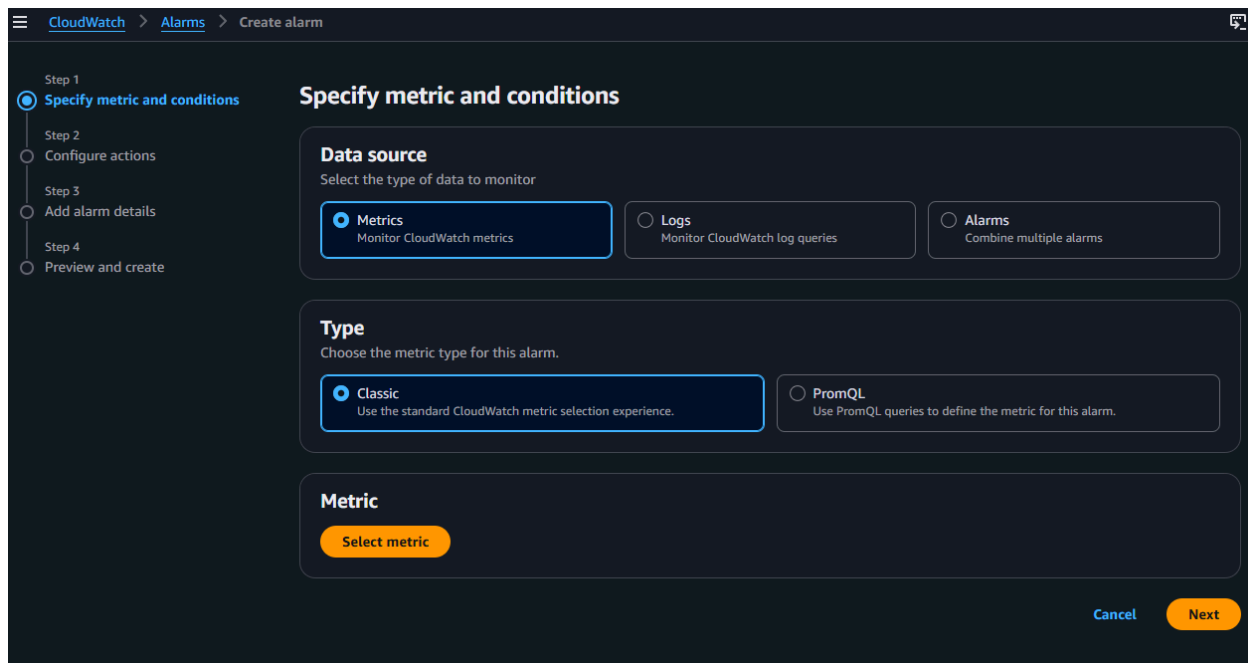
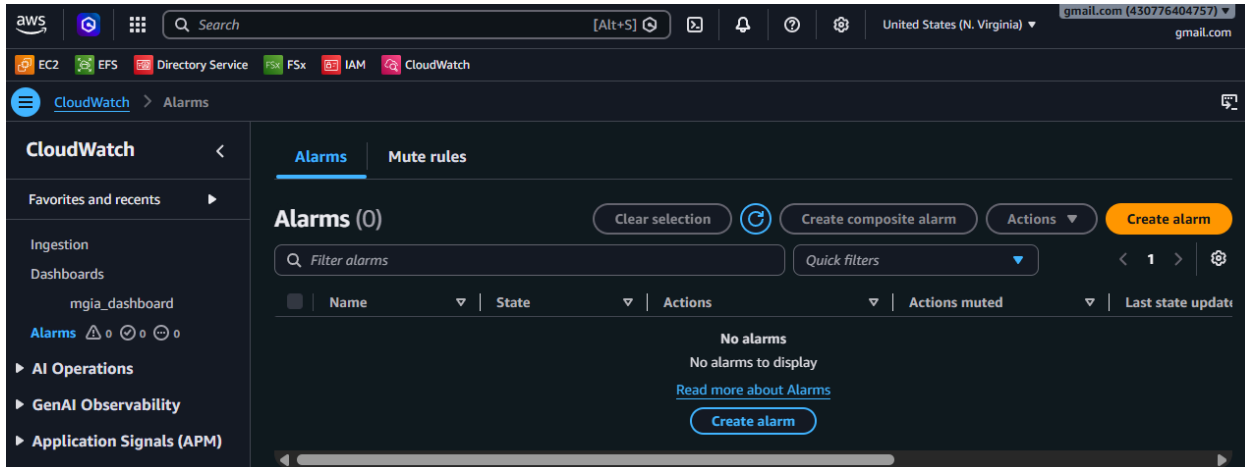
3. Under **Alert preferences**, enabled **Receive CloudWatch billing alerts**. Saved the preference.



Task 1 — Created Alarm 1 : CloudWatch billing alarm

Step 1: Create alarm

1. Opened Amazon CloudWatch
2. Selected **Alarms** → **Clicked Create alarm.**
3. **Selected the Billing Metric for Alarm 1**
4. Clicked **Select metric**



Step 2: Selected the Billing Metric

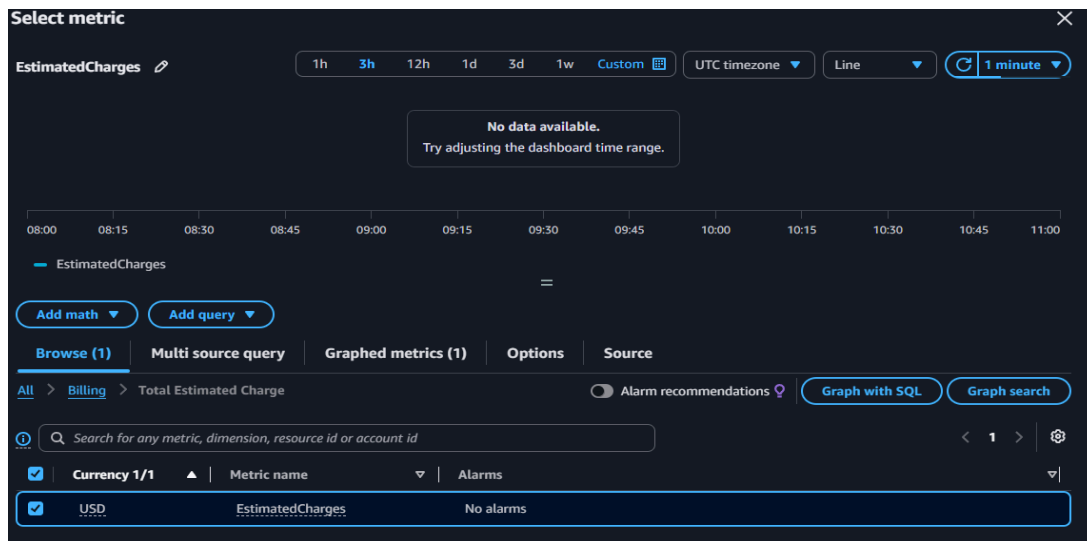
Selected the **Billing** namespace.

Selected **Total Estimated Charge**.

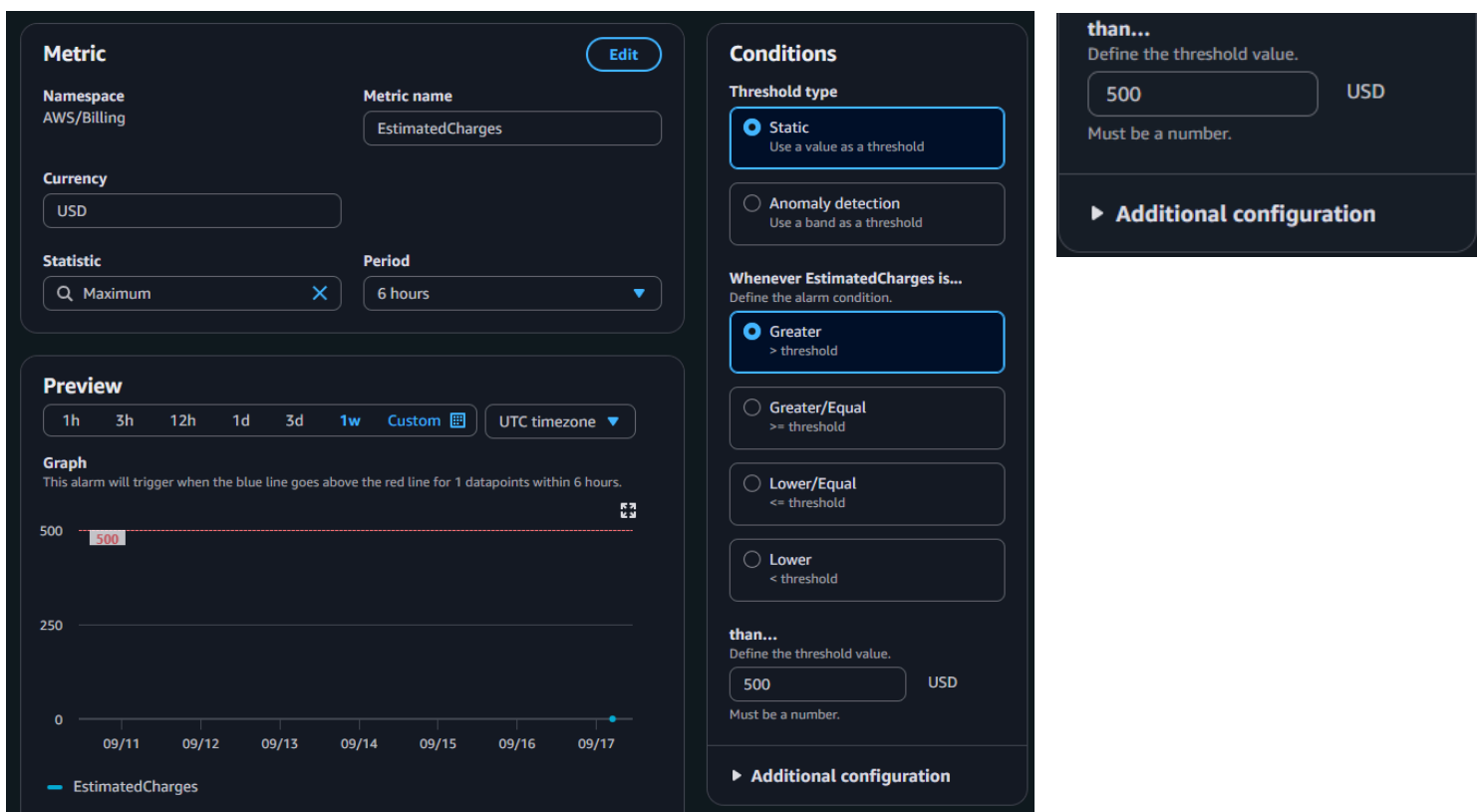
Selected the **EstimatedCharges** metric.

Selected **USD** as the currency.

Selected the metric.



Step 3: Configured the \$500 Threshold



Step 4: Configured SNS Notification

Selected **In alarm** as the alarm state trigger.

Selected **Create a new topic**.

Created SNS topic **Default_CloudWatch_Alarms_Topic**.

Verified that an email endpoint was associated with the SNS topic.

Proceeded to the alarm details.

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☐ Select an existing SNS topic

☒ **Create new topic**

☐ Use topic ARN to notify other accounts

Create a new topic...
The topic name must be unique.

Default_CloudWatch_Alarms_Topic

SNS topic names can contain only alphanumeric characters, hyphens (-) and underscores (_).

Email endpoints that will receive the notification...
Add a comma-separated list of email addresses. Each address will be added as a subscription to the topic above.

mgib.edu@gmail.com

user1@example.com, user2@example.com

Create topic

Add notification

Step 5: Added Alarm Details

Entered the alarm name: **XYZ_Corporation_Billing_Alarm_500**

Reviewed the alarm configuration and Created the billing alarm.



Conditions

Threshold type
Static

Whenever **EstimatedCharges** is
Greater (>)

than...
500

► Additional configuration

Step 2: Configure actions

Edit

Actions

Notification

When In alarm, send a notification to "Default_CloudWatch_Alarms_Topic"

Step 3: Add alarm details

Edit

Alarm details

Name
XYZ_Corporation_Billing_Alarm_500

Description
-

► Tags (0)

CloudWatch > Alarms

CloudWatch

Favorites and recents

Ingestion

Dashboards

Alarms 0 1 0

► AI Operations

► GenAI Observability

► Application Signals (APM)

► Infrastructure Monitoring

► Logs New

► Metrics

► Network Monitoring

► Setup

Observability Workshops 12

Some subscriptions are pending confirmation

Amazon SNS doesn't send messages to an endpoint until the subscription is confirmed

View SNS Subscriptions

Alarms

Mute rules

Alarms (1)

Clear selection

Create composite alarm

Actions

Create alarm

Filter alarms

Quick filters

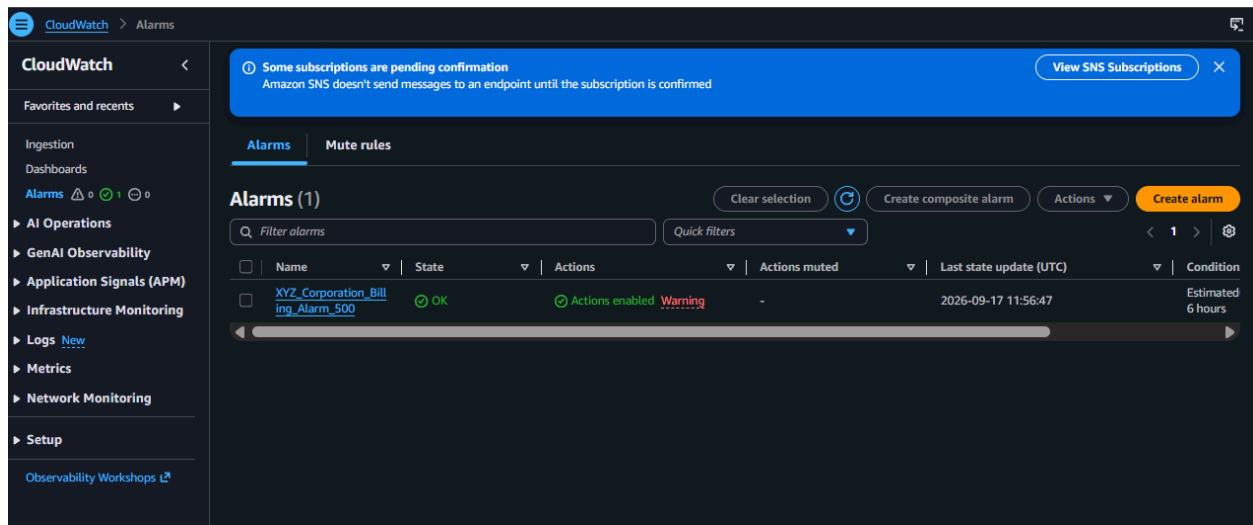
☐	Name	State	Actions	Actions muted	Last state update (UTC)	Condition
☐	XYZ_Corporation_Billing_Alarm_500	OK	Actions enabled	Warning	-	2026-09-17 11:56:47
						Estimated 6 hours

Figure : Create billing alarm : XYZ_Corporation_Billing_Alarm_500

Task 2 — Created Alarm 2: CloudWatch EC2 CPU utilization alarm at 65%

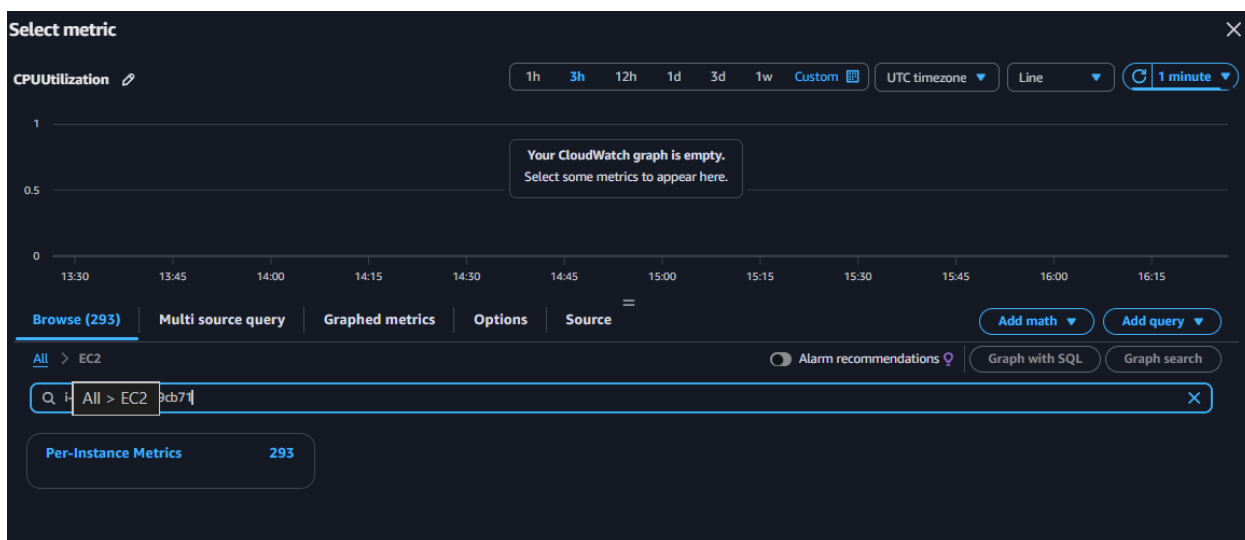
Step 1 — Opened CloudWatch Alarms

1. Again Clicked **Create alarm**.
2. Clicked **Select metric**.



Step 2 — Selected the EC2 Metric

1. Selected **EC2**.
2. Selected **Per-Instance Metrics**.
3. Selected the required EC2 instance **mgia_instance**.
4. Selected the **CPUUtilization** metric.
5. Selected the metric.



[All](#) > [EC2](#) > Per-Instance Metrics

Found 22 matches

	Instance name 22/22	InstanceId	Metric name	Alarms
☐	mgia_instance	i-0e168ccb91829c...	CPUCreditUsage ⓘ	No alarms
☐	mgia_instance	i-0e168ccb91829c...	CPUSurplusCreditBalance ⓘ	No alarms
☐	mgia_instance	i-0e168ccb91829c...	CPUCreditBalance ⓘ	No alarms
☐	mgia_instance	i-0e168ccb91829c...	CPUSurplusCreditsCharged ⓘ	No alarms
☐	mgia_instance	i-0e168ccb91829c...	EBSReadOps ⓘ	No alarms
☐	mgia_instance	i-0e168ccb91829c...	EBSWriteBytes ⓘ	No alarms
☒	mgia_instance	i-0e168ccb91829c...	CPUUtilization ⓘ	No alarms

Step 3 — Configured the 65% CPU Threshold

Condition Greater than 65%

[CloudWatch](#) > [Alarms](#) > Create alarm

☐ Alarm recommendations ⓘ [View details](#)

Step 1: **Specify metric and conditions**
 Step 2: Configure actions
 Step 3: Add alarm details
 Step 4: Preview and create

Specify metric and conditions

Data source
Select the type of data to monitor

☒ **Metrics**
Monitor CloudWatch metrics

☐ **Logs**
Monitor CloudWatch log queries

☐ **Alarms**
Combine multiple alarms

Type
Choose the metric type for this alarm.

☒ **Classic**
Use the standard CloudWatch metric selection experience.

☐ **PromQL**
Use PromQL queries to define the metric for this alarm.

Metric

Namespace: AWS/EC2
 Metric name: CPUUtilization
 InstanceId: i-0e168ccb91829cb71
 Instance name: mgia_instance
 Statistic: Average ⓘ
 Period: 5 minutes

Conditions

Threshold type

☒ **Static**
Use a value as a threshold

☐ **Anomaly detection**
Use a band as a threshold

Whenever CPUUtilization is...
Define the alarm condition.

☒ **Greater**
> threshold

☐ **Greater/Equal**
>= threshold

☐ **Lower/Equal**
<= threshold

☐ **Lower**
< threshold

than...
Define the threshold value.

Must be a number.

Preview

Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

Percent

65%

32.55%

0.11%

13:30 14:00 14:30 15:00 15:30 16:00

CPUUtilization

Step 4 — Configured SNS Notification

Selected **In alarm** as the alarm state trigger.
Selected **Select an existing SNS topic**.
Selected **Default_CloudWatch_Alarms_Topic**.
Verified the associated email endpoint.

The screenshot shows the 'Configure actions' step in the AWS CloudWatch console. The left sidebar indicates the current step is 'Configure actions'. The main panel is titled 'Configure actions' and contains a 'Notification' section. Under 'Alarm state trigger', the 'In alarm' option is selected. Below this, the 'Send a notification to the following SNS topic' section has 'Select an existing SNS topic' chosen. A search bar labeled 'Select an SNS topic' is visible, and the 'Add notification' button is at the bottom.

Step 5 — Added Alarm Details

1. Entered the alarm name: **XYZ_Corporation_EC2_CPU_Alarm_65**
2. Reviewed the configuration.
3. Created the EC2 CPU alarm.

The screenshot shows the 'Add alarm details' step in the AWS CloudWatch console. The left sidebar indicates the current step is 'Add alarm details'. The main panel is titled 'Add alarm details' and contains a 'Name and description' section. The 'Alarm name' field is filled with 'XYZ_Corporation_EC2_CPU_Alarm_65'. The 'Alarm description - optional' section is expanded, showing a text area with the description 'Alarm triggers when EC2 CPU utilization exceeds 65%'. Below this, there is a note about markdown formatting. The 'Tags - optional' section is also visible, showing 'No tags associated with the resource' and an 'Add new tag' button. At the bottom right, there are 'Cancel', 'Previous', and 'Next' buttons.

Step 1

Specify metric and conditions

Step 2

Configure actions

Step 3

Add alarm details

Step 4

Preview and create

Preview and create

Step 1: Specify metric and conditions

Metric

Namespace

AWS/EC2

Metric name

CPUUtilization

Instanceld

i-0e168ccb91829cb71

Instance name

mgia_instance

Statistic

Average

Period

5 minutes

Preview

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

Percent

65%

32.55%

0.11%

13:30 13:45 14:00 14:15 14:30 14:45 15:00 15:15 15:30 15:45 16:00 16:15 16:30

CPUUtilization

Conditions

Threshold type

Static

Whenever CPUUtilization is

Greater (>)

than...

65

Additional configuration

Step 2: Configure actions

Actions

Notification

When In alarm, send a notification to "Default_CloudWatch_Alarms_Topic"

Step 3: Add alarm details

Alarm details

Name

XYZ_Corporation_EC2_CPU_Alarm_65

Description

Alarm triggers when EC2 CPU utilization exceeds 65%.

Tags (0)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Cancel

Previous

Create alarm

Results : Successfully created and configured CloudWatch billing and EC2 CPU alarms with SNS notifications, both showing OK status.

CloudWatch > Alarms

Alarms

Mute rules

Alarms (2)

Filter alarms

Quick filters

Clear selection

Create composite alarm

Actions

Create alarm

☐	Name	State	Actions	Actions muted	Last state update (UTC)	Conditions
☐	XYZ_Corporation_EC2_CPU_Alarm_65	OK	Actions enabled Warning	-	2026-09-17 16:33:10	CPUUtilization > 65 for 1 datapoints within 5 minutes
☐	XYZ_Corporation_Billing_Alarm_500	OK	Actions enabled Warning	-	2026-09-17 11:56:47	EstimatedCharges > 500 for 1 datapoints within 6 hours